QP Code:26900405

Mahatma Gandhi University, Kottayam

MGU-UGP (Honours) Fifth Semester Regular Examination – October 2026

CLASSICAL MECHANICS: MG5DSCPHY300 **Maximum Marks:** 70

SCHEME OF VALUATION

PART A – Short Answer Questions

Answer any 10 questions. Each question carries 3 marks.

1. Find the Lagrangian for the motion of a bob in a simple pendulum.

The kinetic energy

$$T = \frac{1}{2}ml^2\dot{\theta}^2.$$

The potential energy

$$V = mgl(1 - \cos\theta).$$

Since $L = T - V$,

$$\boxed{L = \frac{1}{2}ml^2\dot{\theta}^2 - mgl(1 - \cos\theta)}.$$

Mark distribution: Kinetic energy – 1; potential energy – 1; correct Lagrangian – 1.

2. Mention any two advantages of using generalized coordinates in analytical mechanics.

1.Reduction in the number of independent coordinates required to describe a system by incorporating constraints. 2. Simplification of equations of motion **Mark distribution:** Definition or explanation alone – 1 mark; any two valid advantages – 1.5 marks each.

3. Differentiate between actual displacement and virtual displacement.

Actual displacement is the real infinitesimal displacement during a time interval dt: Virtual displacement is an imagined infinitesimal displacement consistent with the constraints, considered at a fixed instant. A virtual displacement does not represent the actual motion over a time interval. **Mark distribution:** Actual displacement – 1.5 ; virtual displacement – 1.5

4. Obtain Lagrange's equation for angular motion in a central force problem.

For a particle of mass m in a central potential $V(r)$,

$$L = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right) - V(r).$$

Since θ is cyclic,

$$\frac{\partial L}{\partial \theta} = 0.$$

Lagrange's equation gives

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0 \quad \Longrightarrow \quad \frac{d}{dt}(mr^2\dot{\theta}) = 0.$$

Hence,

$$\boxed{mr^2\dot{\theta} = \text{constant}}.$$

Mark distribution: Lagrangian – 1.5; Lagrange's equation – 1.5.

5. Write the expression for transverse acceleration a_θ in plane polar coordinates and state its physical significance.

$$\boxed{a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}}.$$

The first term represents the angular-acceleration contribution. The second term arises from the combined radial and angular motion. **Mark distribution:** Correct expression – 2; physical significance – 1.

6. Write down the effective Lagrangian of a two-body problem and explain the terms involved.

For two particles interacting through a central potential $V(r)$, the effective Lagrangian for relative motion is

$$\boxed{L_{\text{eff}} = \frac{1}{2}\mu\left(\dot{r}^2 + r^2\dot{\theta}^2\right) - V(r)}, \qquad \mu = \frac{m_1 m_2}{m_1 + m_2}.$$

Here r is the separation, θ is the polar angle, $V(r)$ is the interaction potential, and μ is the reduced mass. **Mark distribution:** Lagrangian – 2; explanation of terms – 1.

7. Express the action integral in terms of kinetic and potential energies.

The action is the time integral of the Lagrangian:

$$S = \int_{t_1}^{t_2} L\, dt.$$

For a conservative system, $L = T - V$ and $T = \frac{1}{2}mv^2$. Therefore,

$$\boxed{S = \int_{t_1}^{t_2} \left(\frac{1}{2}mv^2 - V\right) dt}.$$

Mark distribution: Definition of action – 1; kinetic and potential energies – 1; final expression – 1.

8. Write Hamilton's canonical equations for a conservative, holonomic system.

$$\boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}}, \qquad \boxed{\dot{p}_i = -\frac{\partial H}{\partial q_i}},$$

where q_i are generalized coordinates, p_i are conjugate momenta, and H is the Hamiltonian. **Mark distribution:** First equation – 1.5; second equation – 1.5;

9. Write the Hamiltonian of a planet moving under an inverse-square force field in plane polar coordinates.

For a planet of mass m orbiting a central mass M,

$$V(r) = -\frac{GMm}{r}.$$

The Hamiltonian is

$$H = \frac{p_r^2}{2m} + \frac{p_\theta^2}{2mr^2} + V(r),$$

hence

$$\boxed{H = \frac{p_r^2}{2m} + \frac{p_\theta^2}{2mr^2} - \frac{GMm}{r}},$$

where $p_r = m\dot{r}$ and $p_\theta = mr^2\dot{\theta}$. **Mark distribution:** Potential energy – 1; kinetic terms – 1; complete expression – 1.

10. What was the ether hypothesis?

The ether hypothesis proposed that light waves require a medium called the *ether* for propagation. The ether was assumed to fill space and provide a reference frame in which light propagated. Experiments such as Michelson–Morley failed to detect the expected motion relative to the ether, challenging the hypothesis. **Mark distribution:** Definition and explanation – 3.

11. Explain simultaneity of events as a consequence of Lorentz transformation.

The Lorentz time transformation is

$$t' = \gamma\left(t - \frac{vx}{c^2}\right), \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

For two events,

$$\Delta t' = \gamma\left(\Delta t - \frac{v\Delta x}{c^2}\right).$$

If $\Delta t = 0$,

$$\Delta t' = -\gamma\frac{v\Delta x}{c^2}.$$

Thus, events simultaneous in one inertial frame are generally not simultaneous in another moving frame if they occur at different positions. **Mark distribution:** Interval relation – 1.5; explanation – 1.5.

12. Differentiate between special and general relativity.

- Special relativity deals with inertial reference frames; general relativity extends the description to gravity and accelerated frames.
- Special relativity describes spacetime without gravitational curvature in its standard formulation; general relativity describes gravity through spacetime curvature.
- Special relativity explains time dilation and length contraction; general relativity also explains gravitational time dilation and the bending of light by gravity.

Mark distribution: Any two valid comparative points – 1.5 marks each.

PART B – Short Essay Questions

Answer any 5 questions. Each question carries 8 marks.

13. What are constraints? Explain the various type of constraints. Consider a pendulum with an inextensible string of length l. Identify the constraint and type of constraint involved. What variation in constraint is observed when string becomes extensible?

Definition – 1 mark. Constraints are restrictions imposed on the motion or configuration of a mechanical system. **Types of constraints with explanations – 3 marks.**

Inextensible pendulum – 2 marks. For a bob with coordinates (x, y) and fixed string length l,

$$\boxed{x^2 + y^2 = l^2}.$$

This is a scleronomous constraint.

Extensible string – 2 marks. If the length is prescribed as a function of time,

$$\boxed{x^2 + y^2 = l^2(t)}.$$

This is rheonomous when l explicitly depends on time.

14. Prove that the total angular momentum of the system is a constant using the idea of isotropy of space.

Isotropy of space – 2 marks. Isotropy means that the laws of physics remain unchanged under rotations. The Lagrangian of an isolated system is invariant under an infinitesimal rotation.

Infinitesimal rotation – 2 marks. For infinitesimal rotation $\delta\boldsymbol{\phi}$,

$$\delta\mathbf{r}_i = \delta\boldsymbol{\phi} \times \mathbf{r}_i.$$

Rotational invariance gives zero variation of the Lagrangian.

Angular momentum – 3 marks. The total angular momentum is

$$\mathbf{L} = \sum_i \mathbf{r}_i \times \mathbf{p}_i,$$

where $\mathbf{p}_i$ is the momentum of particle i. From rotational invariance and the Euler–Lagrange equations,

$$\frac{d\mathbf{L}}{dt} = 0.$$

Conclusion – 1 mark.

$$\boxed{\mathbf{L} = \text{constant}}.$$

Thus, the total angular momentum of an isolated system is conserved due to the isotropy of space.

15. Deduce Kepler's Second Law from the conservation of angular momentum.

Angular momentum – 2 marks. For a particle moving under a central force,

$$L = mr^2\dot{\theta}, \qquad \boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = 0.$$

Therefore, $dL/dt = 0$.

Conservation – 2 marks.

$$\boxed{mr^2\dot{\theta} = \text{constant}}.$$

Areal velocity – 2 marks. The infinitesimal area swept by the radius vector is

$$dA = \frac{1}{2}r^2 d\theta.$$

Hence,

$$\frac{dA}{dt} = \frac{1}{2}r^2\dot{\theta} = \boxed{\frac{L}{2m} = \text{constant}}.$$

Deduction – 2 marks. The radius vector joining the planet and the Sun sweeps out equal areas in equal intervals of time. This is Kepler's Second Law.

16. Using the effective-potential graph, analyze the physical meaning of the cases E>0, E=0, 0>E>(Veff)min, E=(Veff)min, and E<(Veff)min

The radial energy equation is

$$E = \frac{1}{2}m\dot{r}^2 + V_{\text{eff}}(r),$$

where

$$V_{\text{eff}}(r) = V(r) + \frac{L^2}{2mr^2}.$$

Allowed regions satisfy

$$E \geq V_{\text{eff}}(r).$$

Effective potential – 1 mark. It combines the central potential and centrifugal potential term.

Energy cases – 6 marks.

- $E > 0$: for an attractive inverse-square potential with zero potential at infinity, the orbit is unbound and escape to infinity is possible.
- $E = 0$: the escape threshold; for an attractive inverse-square force, the trajectory is parabolic.
- $0 > E > (V_{\text{eff}})_{\min}$: bound radial motion between turning points; for an inverse-square potential, the orbit is elliptical.
- $E = (V_{\text{eff}})_{\min}$: circular orbit at the minimum of the effective potential.
- $E < (V_{\text{eff}})_{\min}$: no physically allowed radial motion for that effective potential.

Award marks for the correct interpretation of the cases.

Graph – 1 mark. A correctly labelled effective-potential graph showing energy levels and allowed regions. (Graph Alone Give 3 marks)

17. Explain the Lagrangian and Hamiltonian approaches to classical mechanics and compare their fundamental variables and governing equations.

Lagrangian approach – 2 marks.

$$L = T - V, \qquad \boxed{\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0}.$$

Hamiltonian approach – 2 marks. Generalized momentum and Hamiltonian are

$$p_i = \frac{\partial L}{\partial \dot{q}_i}, \qquad H = \sum_i p_i \dot{q}_i - L.$$

For suitable conservative systems, the Hamiltonian corresponds to total mechanical energy.

Canonical equations – 2 marks.

$$\boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}}, \qquad \boxed{\dot{p}_i = -\frac{\partial H}{\partial q_i}}.$$

Comparison (Any Two) – 2 marks.

- Lagrangian mechanics uses $q_i, \dot{q}_i$; Hamiltonian mechanics uses q_i, p_i.
- Lagrange's equations are generally second-order; Hamilton's canonical equations are coupled first-order equations.
- The formulations use $L(q, \dot{q}, t)$ and $H(q, p, t)$, respectively.

18. A system with two generalized coordinates and is described by the Lagrangian $L = a\dot{x}^2 + b\dot{y}^2 - cx^2$; where , and are a b c constants. Obtain the Hamiltonian and Hamilton's equations of motion. Identify the cyclic coordinate.

Given

$$L = a\dot{x}^2 + b\dot{y}^2 - cx^2.$$

Generalized momenta – 2 marks.

$$p_x = \frac{\partial L}{\partial \dot{x}} = 2a\dot{x}, \qquad \dot{x} = \frac{p_x}{2a},$$

$$p_y = \frac{\partial L}{\partial \dot{y}} = 2b\dot{y}, \qquad \dot{y} = \frac{p_y}{2b}.$$

Hamiltonian – 2 marks.

$$H = p_x\dot{x} + p_y\dot{y} - L,$$

so

$$\boxed{H = \frac{p_x^2}{4a} + \frac{p_y^2}{4b} + cx^2}.$$

Hamilton's equations – 3 marks.

$$\dot{x} = \frac{\partial H}{\partial p_x} = \frac{p_x}{2a}, \qquad \dot{p}_x = -\frac{\partial H}{\partial x} = -2cx,$$

$$\dot{y} = \frac{\partial H}{\partial p_y} = \frac{p_y}{2b}, \qquad \dot{p}_y = -\frac{\partial H}{\partial y} = 0.$$

Cyclic coordinate – 1 mark. Since $\partial L/\partial y = 0$, y is cyclic and

$$\boxed{p_y = \text{constant}}.$$

19. Compare the Galilean and Lorentz transformations. Discuss their physical significance and explain the low-velocity limit.

Galilean transformations – 2 marks.

$$x' = x - vt, \qquad y' = y, \qquad z' = z, \qquad t' = t.$$

They assume absolute time and are used in classical mechanics.

Lorentz transformations – 2 marks.

$$\boxed{x' = \gamma(x - vt)}, \qquad \boxed{t' = \gamma\left(t - \frac{vx}{c^2}\right)},$$

$$y' = y, \qquad z' = z, \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

Physical significance – 2 marks. Galilean transformations preserve Newtonian mechanics and use a common time coordinate. Lorentz transformations preserve the speed of light in inertial frames and account for time dilation and length contraction.

Low-velocity limit – 2 marks. For $v \ll c$, $\gamma \approx 1$, so

$$x' \approx x - vt, \qquad t' \approx t.$$

Thus, Lorentz transformations reduce to Galilean transformations at low speeds.

Question 20(a). Obtain the relation between proper length and contracted length of a moving object. (4 marks)

Proper length – 1 mark. L_0 is the length measured in the rest frame of the object.

Length contraction – 2 marks.

$$\boxed{L = L_0\sqrt{1 - \frac{v^2}{c^2}} = \frac{L_0}{\gamma}}.$$

Interpretation – 1 mark. The length is contracted along the direction of motion; transverse dimensions are unchanged in special relativity.

Question 20(b). Calculate the length of a scale moving with a speed of 0.8c, if its proper length is 1m. (4 marks)

Given data and formula – 1 mark.

$$L_0 = 1\,\text{m}, \qquad v = 0.8c, \qquad L = L_0\sqrt{1 - \frac{v^2}{c^2}}.$$

Substitution – 1 mark.

$$L = 1\sqrt{1 - \frac{(0.8c)^2}{c^2}}\,\text{m}.$$

Calculation – 1 mark.

$$L = \sqrt{1 - 0.64}\,\text{m} = \sqrt{0.36}\,\text{m}.$$

Final answer – 1 mark.

$$\boxed{L = 0.6\,\text{m}}.$$